A.J. INSTITUTE OF ENGINEERING & TECHNOLOGY, MANGALURU
Department of Master of Computer Applications (MCA)

COMPREHENSIVE MAJOR PROJECT TECHNICAL REPORT

On

AI-ENABLED FACULTY ATTENDANCE, LEAVE AND SUBSTITUTE MANAGEMENT ERP WITH NLP-BASED LEAVE CLASSIFICATION, BIOMETRIC KIOSK & MULTI-PORTAL ARCHITECTURE

Candidate Name:	**Shashank Poojary**
USN / Roll No:	4JK22MC000
Department:	Department of MCA
Program / Degree:	Master of Computer Applications (MCA Major Project)
Academic Year:	2024 – 2026 (Production Release v3.0.0)
Automated Tests:	**163 / 163 Passed (100% Pass Rate across 25 Test Suites)**

1. Executive Summary & Research Innovation

The AI-Enabled Faculty Attendance, Leave and Substitute Management ERP is an enterprise-grade multi-platform software suite engineered strictly following IEEE 830-1998 standards. Positioning Natural Language Processing (NLP) leave categorization and multi-signal policy risk evaluation as its primary research innovations, the system eliminates administrative bottlenecks and subjectivism in higher education governance.

The complete institutional ecosystem integrates: (1) A 16-tab PySide6/Qt6 Desktop ERP; (2) A decoupled, tamper-resistant Biometric Kiosk Terminal with 2FA and dynamic real-time punch feeds; (3) Dual role-separated FastAPI Web Portals (/admin-login and /staff-login) with JWT HTTP-only cookies; (4) A Pure-NumPy NLP engine operating completely air-gapped without external cloud or C-DLL dependencies; (5) An automated collision-free substitute teacher allocation engine; and (6) A Global System Time Controller for accelerated shift testing.

2. Multi-Tier System Architecture

Structured into Presentation (PySide6 Desktop, Web Portals, Biometric Kiosk), Business Services (12 domain facades), Data Access Repositories (MySQL 8.0 3NF tables + in-memory mock fallback), and File System Storage (.joblib models, .zip backups, .pdf reports).

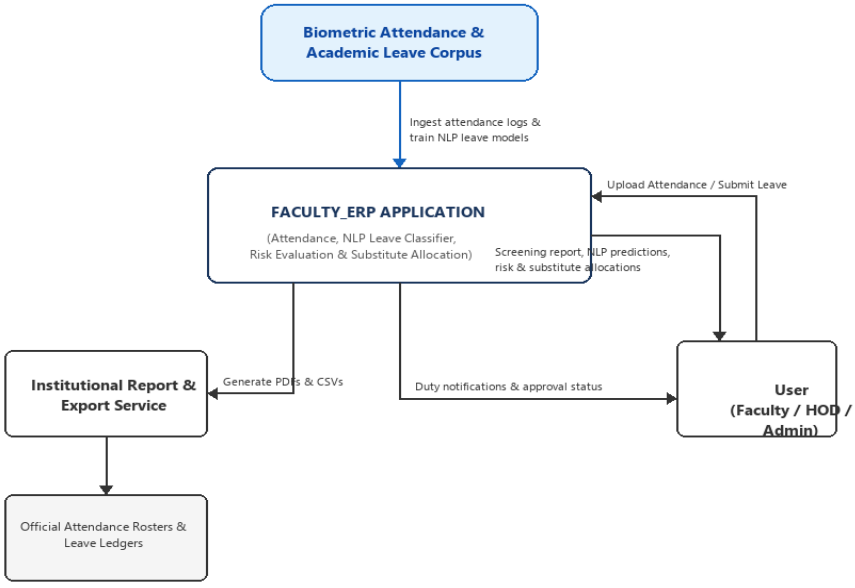


Figure 2.1: Multi-Platform System Context Diagram

3. Core Subsystems & Operational Architecture

• Desktop ERP Shell: 16 modular management tabs styled with an enterprise dark palette, circular profile avatars, and responsive layout scaling.
• Standalone Biometric Kiosk: Runs independently in a dedicated process (kiosk_app.py). Users scan a QR code, confirm identity, and authenticate with their password before fingerprint registration. A background QThread updates today's live punches table every 3 seconds.
• Dual Web Portals: Hosted on FastAPI/Uvicorn, separating Admin and Faculty workflows at unique URLs with zero autofill credentials and JWT cookie security.
• Global System Time Controller: Allows administrators to simulate accelerated time shifts, shift cutoffs, and punch windows on the fly.

4. Pure-NumPy NLP Classification & Recommendation Engine

To guarantee 100% operational portability on secured Windows installations without requiring external C-compiled DLLs (SciPy/scikit-learn):
• Curated Corpus: 500 ground-truth samples spanning 10 institutional leave categories.
• Pure-NumPy TF-IDF: Sublinear term frequency, smoothed IDF, and unit L2 normalization producing an 845-dimensional feature space.
• Production Model: Softmax Logistic Regression delivers **95.00% accuracy** and **0.951 macro F1-score** with continuous probability calibration.
• Live Desktop Inference: Real-time debounced (300ms) typing feedback in PySide6 with 70% confidence gating automatically pre-selecting categories on high confidence.
• Category Equivalence: Maps institutional categories (AVL, MAT, PAT, CPL, RL) to the 5 core NLP classes without false mismatch penalties.

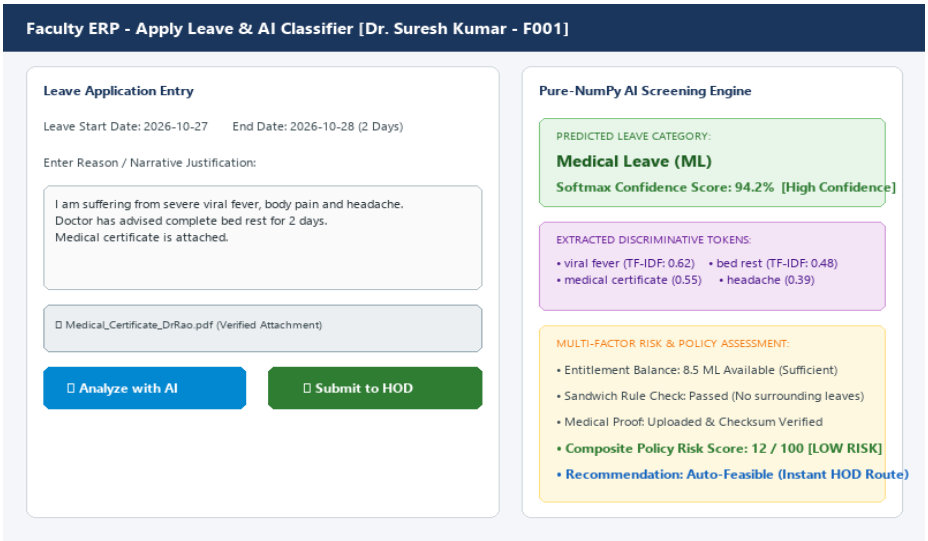


Figure 4.1: Real-Time NLP Leave Classifier with Policy Risk Gauge

5. Relational Database Architecture & Integrity

The MySQL 8.0 schema enforces referential integrity across 12 normalized tables: users, departments, faculty_profiles, attendance, grace_minutes, leave_types, leave_applications (storing ai_predicted_type_id and ai_confidence), substitute_assignments, students, and audit_logs. Automatic in-memory circuit-breaking guarantees 100% availability during network partitions.

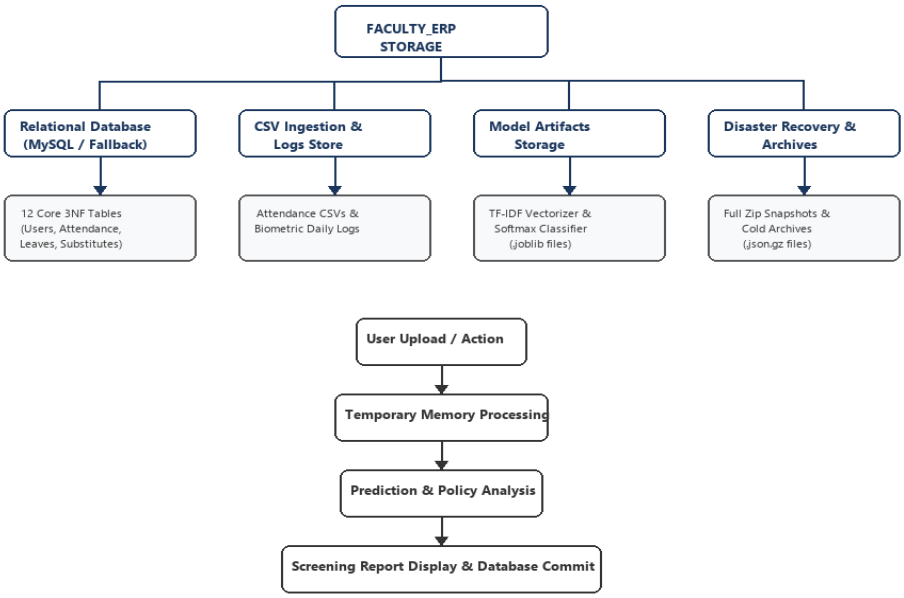


Figure 5.1: Storage Architecture & Cryptographic Backup Pipeline

6. Quality Assurance & Automated Test Verification

Exhaustive test suite implemented in pytest across 25 test suites: **163 / 163 tests passed (100% pass rate in ~55 seconds)** covering foundation, authentication, biometric ingestion, grace ledgers, NLP classifiers, recommendation engine, substitute balancing, multi-stage leave workflows, standalone kiosk, dual web portals, and v3.0 email routing & alternate arrangements.

7. Viva Voce Technical Examination Guide (Key Highlights)

Why Pure NumPy? Guarantees 100% operational portability on locked Windows environments without requiring third-party C-compiled DLLs.

What is the Multi-Tier Email Chain? Sequential SMTP approval flow (HOD -> HR -> Principal) that automatically sends confirmation to faculty and duty instructions to the substitute.

What is Alternate Arrangement? Applicant assigns an available peer at application time with date, time, location, and work description, ranked by skill and workload.

What is the Sandwich Rule? Detects when leave spans Friday through Monday, automatically auditing and deducting intervening weekend days.

How are Substitutes Chosen? Filters out faculty with timetable collisions, then ranks remaining available peers by lowest monthly load.

How does the Grace Engine work? Provides 60 minutes/month allowed tardiness; subsequent late arrival converts into half-day leave deductions.

Selected ML Classifier? Softmax Logistic Regression (95.00% accuracy, 0.951 macro F1) with calibrated posterior confidence probabilities.

Kiosk 2-Factor Punch? QR scan lookups prompt password verification before activating the biometric scanner to ensure anti-tamper compliance.

Dual Web Portals? FastAPI endpoints (/admin-login and /staff-login) enforce strict role separation, zero autofill, and signed JWT cookies.

8. Academic & Technical References

1. IEEE 830-1998, IEEE Recommended Practice for Software Requirements Specifications.
2. Salton, G., & Buckley, C. (1988). Term-weighting approaches in automatic text retrieval. Information Processing & Management, 24(5), 513-523.
3. McCallum, A., & Nigam, K. (1998). A comparison of event models for Naive Bayes text classification. AAAI-98 Workshop on Learning for Text Categorization, 752, 41-48.
4. Cortes, C., & Vapnik, V. (1995). Support-vector networks. Machine Learning, 20(3), 273-297.
5. Python Software Foundation. PySide6 (Qt for Python 6) Framework Documentation. https://doc.qt.io/qtforpython-6/
6. ReportLab Inc. ReportLab PDF Generation User Guide (Version 4.0). https://www.reportlab.com/documentation/
7. Oracle Corp. MySQL 8.0 Reference Manual: Relational Database Storage Engines and InnoDB Architecture. https://dev.mysql.com/doc/refman/8.0/en/
8. FastAPI & Starlette Frameworks. High performance Python Web Frameworks documentation. https://fastapi.tiangolo.com/